

Approved rendering: evaluation 2

Surprisal and DINO rank outcomes best here; extra history and fusion show no consistent gain. Better rendering helps DT, but arm-pose-sensitive completion thresholds remain the main failure.

Two central plots: ranking.png and decisions.png. Every PSI/DINO threshold and logistic fit is unchanged; Robometer success uses native probability ≥ 0.5 , while its progress head is used for ranking only.

Method	50 original BA	50 goal-pose BA	70 original BA	70 goal-pose BA	90 original recall	90 goal-pose recall
DT	0.938	0.500	0.929	0.500	0.917	1.000
S1	1.000	0.500	0.900	0.500	0.333	1.000
S3	1.000	0.500	0.900	0.500	0.333	1.000
S5	1.000	0.500	0.900	0.500	0.333	1.000
DT + S1	1.000	0.500	0.900	0.500	0.333	1.000
DT + S3	1.000	0.500	0.900	0.500	0.333	1.000
DT + S5	1.000	0.500	0.900	0.500	0.333	1.000
DINOv2	1.000	0.500	0.900	0.500	0.333	1.000
Robometer progress	N/A	N/A	N/A	N/A	N/A	N/A
Robometer success	0.500	0.500	0.500	0.500	0.083	0.000

Limitations:

- Four reused test scenes; related observations across six conditions are not independent.
- Renderer and resolution change together; cannot attribute their effects separately.
- 90% branches are all successful: no outcome AUROC/BA, only success recall.
- Common-pose condition adds 10 seconds of physics without an equal-duration no-motion control.
- PSI/DINO use matched successful visual goals; Robometer uses task language and eight prefix frames.
- Robometer is newly scored on this suite; no old-render Robometer comparison or dev-fitted completion threshold.
- Old PSI/DINO thresholds and fusion remain frozen; this measures threshold transfer across rendering, not best achievable recalibration.

Full protocol and source-cluster uncertainty

```
{
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    "version": "synthetic_mugs_native256_v4",
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        "source_frames": 974,
        "branch_frame": 486,
        "branch_completion": 0.5,
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        "initial_heights": [
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        "mug_addresses": [
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        ],
        "split": "dev",
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          "scene_qpos.npy": "681134f39a5d9b072f27f2bca566dcfa49ab88fa4b81a90db2f642bd66629af0",
          "states_actions.bin": "8287fa068ec796635d9e2c494011da256851c596ff0e5f3d37000815bcbd622e",
          "randomization.json": "2b64f59c34654455bcc22ff6e0868f0847b0dec82150dd106583902d87f21d59"
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  }
}
```

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      "states_actions.bin": "5c8344d8b894dd54d62fe0f67e55000a2a5c3ec8f40e2c6da500f28b8674f6c9",
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```

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  "states_actions.bin": "840698fdb1fce3dc64baaf8349cd3101424f88ec305c9c8993090dfef4fc1a14",
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  "branch_frame": 413,
  "branch_completion": 0.5,
  "source_final_completion": 1.0,
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    0.808255532455444
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    "scene_qpos.npy": "00d4cc5babee09e7681da3eef9c6422ab310a61c55eb3432942fe2194bb550dd",
    "states_actions.bin": "925d141a20c67c2ce4c2e3c990187cb5c7fcc3af4990508f0e409d8ea0073bd5",
    "randomization.json": "76a76d91e7adbb71f72fe66d67e13ef19fd276658be09d18428f6577e1bae95c"
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  "branch_frame": 327,
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  "mug_addresses": [
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}
],
"cells": [
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    "path":
"/ccn2b/dataset/brd/rwm_progress_tracking/synthetic_mugs_native256_v4/branch_50_original",
    "old_path": "/ccn2b/dataset/brd/rwm_progress_tracking/synthetic_mugs_v1"
  }
]

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      "arm_condition": "parked",
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      "branch_percent": 70,
      "arm_condition": "original",
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      "old_path":
"/ccn2b/dataset/brd/rwm_progress_tracking/synthetic_mugs_difficulty_v3/branch_70_original"
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      "arm_condition": "original",
      "path":
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      "old_path":
"/ccn2b/dataset/brd/rwm_progress_tracking/synthetic_mugs_difficulty_v3/branch_90_original"
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    {
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      "path": "/ccn2b/dataset/brd/rwm_progress_tracking/synthetic_mugs_native256_v4/branch_90_parked",
      "old_path":
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  ],
  "design": "Same 108 terminal conditions / six source scenes. Rerender every observation and goal; do
not reuse old image codes or model scores.",
  "branch_times": "floor((source_frames - 1) * fraction) at 50%,70%,90% of recorded duration. Time
fraction is not task completion. No source replacement or outcome-based filtering.",
  "arm_conditions": "Original generated endpoint versus 10-second physical motion to the successful
source terminal robot pose. Objects not edited. Actual labels re-evaluated after motion.",
  "modes": [
    "replay_control",
    "hold",
    "random_walk"
  ],
  "thresholds": "/ccn2b/dataset/brd/rwm_progress_tracking/synthetic_mugs_v1/threshold_metrics.json",
  "thresholds_sha256": "f277e521c2c662cb92be6566343e869593e31e41f9c7451631aff7c1e2661062",
  "evaluation": "Original development thresholds and logistic coefficients fixed. Reuse original
source-group split (2 dev,4 test); report each cell separately, source-cluster uncertainty, no claim of
fresh independent generalization.",
  "outcomes": "Use measured ABC completion, retain successes among perturbed branches and changes
during parking. Single-class cells have undefined AUR0C and balanced accuracy.",
  "generation_status": "Frozen before added 70/90% outcomes or model scores",
  "limits": [
    "Late branches may already be successful.",
    "Only six scenes; 108 observations are not 108 independent scenes."
  ]

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    "This design was motivated by results on the same original scenes.",
    "Retrospective successful demonstration goals."
  ],
  "previous_render_run": "/ccn2b/dataset/brd/rwm_progress_tracking/synthetic_mugs_difficulty_v3",
  "user_approval": "User confirmed the 342x256 native MuJoCo preview and requested rerunning model
ablations and baselines.",
  "rendering": {
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    "width": 342,
    "height": 256,
    "camera": "top",
    "lighting": "saved scene lights/materials",
    "antialiasing": "saved 16x MSAA",
    "geometry_states": "unchanged",
    "approval_preview": "/ccn2b/dataset/brd/rwm_progress_tracking/render_appearance_preview_v1"
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  "psi_preprocessing": "Keep existing audited resize to H256 x W336 with antialiasing, normalized
image tokenizer, seeded 128 patches/frame; same checkpoint and DT/S1/S3/S5 recipes.",
  "robometer_protocol": "New synthetic baseline: pinned Robometer-4B, task language and eight
uniformly sampled causal-prefix images, including initial and terminal frames. Native progress AUROC;
native success-head threshold 0.5. No fitted Robometer threshold and no previous-render Robometer result
claimed.",
  "threshold_policy": "Existing PSI/DINO thresholds and logistic fits remain frozen from the original
development run. Robometer success head uses native 0.5; progress head has AUROC only.",
  "original_artifacts_sha256": {
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    "scoring_manifest.json": "880598865ae53b61abcalcel60465711008d2247f359aeb11db0ce57d3b226a6",
    "psi_scores.jsonl": "45a381d242c7671e066ea19d12c4c5c5a15a5a9a73a1258c367aa9445460eddc",
    "dino_scores.json": "e52a700fe7bb81de17a12112d96ce73e5738ad344acacee25d7e9baae3a6400c",
    "difficulty_metrics.json": "4bfff75d4ad95a5b5b218eda5e4c002a7811261e9805d8337378b3ad62bab5e7c"
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      ],
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        1.0
      ],
      "recall_success": [
        1.0,

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{
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}
},
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}
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    ],
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    1.0
  ],
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      1.0
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  ],
}

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        1.0
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        1.0
      ],
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        1.0
      ],
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        1.0,
        1.0
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  {
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    "arm_condition": "original",
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    "tp": 4,
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      "auroc": [
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  },

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},
{
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  "recall_incomplete": null,
  "tn": null,
  "fp": null,
  "fn": null,
  "tp": null,
  "within_source_auroc": 0.125,
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    "balanced_accuracy": null,
    "recall_success": null,
    "recall_incomplete": null
  }
},
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    0.5
  ],
  "recall_success": [
    0.0,
    0.0
  ],
  "recall_incomplete": [
    1.0,
    1.0
  ]
}
},
{
  "branch_percent": 50,
  "arm_condition": "parked",
  "method": "DINOv2",
  "threshold": 0.8599722981452942,
  "auroc": 1.0,
  "balanced_accuracy": 0.5,
  "accuracy": 0.3333333333333333,
  "recall_success": 1.0,
  "recall_incomplete": 0.0,
  "tn": 0,
  "fp": 8,
  "fn": 0,
  "tp": 4,
  "within_source_auroc": 1.0,
  "ci95_source_cluster": {
    "auroc": [
      1.0,
      1.0
    ],
    "balanced_accuracy": [
      0.5,
      0.5
    ],
    "recall_success": [
      1.0,
      1.0
    ],
    "recall_incomplete": [
      0.0,
      0.0
    ]
  }
}
},
{
  "branch_percent": 50,
  "arm_condition": "parked",
  "method": "DT",
  "threshold": -8.942145824432373,
  "auroc": 1.0,
  "balanced_accuracy": 0.5,
  "accuracy": 0.3333333333333333,
  "recall_success": 1.0,
  "recall_incomplete": 0.0,
  "tn": 0,
  "fp": 8,
  "fn": 0,

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"tp": 4,
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"ci95_source_cluster": {
  "auroc": [
    1.0,
    1.0
  ],
  "balanced_accuracy": [
    0.5,
    0.5
  ],
  "recall_success": [
    1.0,
    1.0
  ],
  "recall_incomplete": [
    0.0,
    0.0
  ]
}
},
{
  "branch_percent": 50,
  "arm_condition": "parked",
  "method": "Elapsed time",
  "threshold": 34.43333333333334,
  "auroc": 0.5,
  "balanced_accuracy": 0.5,
  "accuracy": 0.4166666666666667,
  "recall_success": 0.75,
  "recall_incomplete": 0.25,
  "tn": 2,
  "fp": 6,
  "fn": 1,
  "tp": 3,
  "within_source_auroc": 0.5,
  "ci95_source_cluster": {
    "auroc": [
      0.5,
      0.5
    ],
    "balanced_accuracy": [
      0.5,
      0.5
    ],
    "recall_success": [
      0.25,
      1.0
    ],
    "recall_incomplete": [
      0.0,
      0.75
    ]
  }
},
{
  "branch_percent": 50,
  "arm_condition": "parked",
  "method": "S1",
  "threshold": -3.272519528865814,

```

```

    "auroc": 0.96875,
    "balanced_accuracy": 0.5,
    "accuracy": 0.3333333333333333,
    "recall_success": 1.0,
    "recall_incomplete": 0.0,
    "tn": 0,
    "fp": 8,
    "fn": 0,
    "tp": 4,
    "within_source_auroc": 1.0,
    "ci95_source_cluster": {
      "auroc": [
        0.875,
        1.0
      ],
      "balanced_accuracy": [
        0.5,
        0.5
      ],
      "recall_success": [
        1.0,
        1.0
      ],
      "recall_incomplete": [
        0.0,
        0.0
      ]
    }
  },
  {
    "branch_percent": 50,
    "arm_condition": "parked",
    "method": "S3",
    "threshold": -3.2162272334098816,
    "auroc": 0.96875,
    "balanced_accuracy": 0.5,
    "accuracy": 0.3333333333333333,
    "recall_success": 1.0,
    "recall_incomplete": 0.0,
    "tn": 0,
    "fp": 8,
    "fn": 0,
    "tp": 4,
    "within_source_auroc": 1.0,
    "ci95_source_cluster": {
      "auroc": [
        0.875,
        1.0
      ],
      "balanced_accuracy": [
        0.5,
        0.5
      ],
      "recall_success": [
        1.0,
        1.0
      ],
      "recall_incomplete": [
        0.0,
        0.0
      ]
    }
  }
}

```

```

    ]
  },
  {
    "branch_percent": 50,
    "arm_condition": "parked",
    "method": "S5",
    "threshold": -3.123962104320526,
    "auroc": 1.0,
    "balanced_accuracy": 0.5,
    "accuracy": 0.3333333333333333,
    "recall_success": 1.0,
    "recall_incomplete": 0.0,
    "tn": 0,
    "fp": 8,
    "fn": 0,
    "tp": 4,
    "within_source_auroc": 1.0,
    "ci95_source_cluster": {
      "auroc": [
        1.0,
        1.0
      ],
      "balanced_accuracy": [
        0.5,
        0.5
      ],
      "recall_success": [
        1.0,
        1.0
      ],
      "recall_incomplete": [
        0.0,
        0.0
      ]
    }
  },
  {
    "branch_percent": 50,
    "arm_condition": "parked",
    "method": "DT + S1",
    "threshold": 0.448230025164487,
    "auroc": 1.0,
    "balanced_accuracy": 0.5,
    "accuracy": 0.3333333333333333,
    "recall_success": 1.0,
    "recall_incomplete": 0.0,
    "tn": 0,
    "fp": 8,
    "fn": 0,
    "tp": 4,
    "within_source_auroc": 1.0,
    "ci95_source_cluster": {
      "auroc": [
        1.0,
        1.0
      ],
      "balanced_accuracy": [
        0.5,
        0.5
      ]
    }
  }
}

```

```

    ],
    "recall_success": [
        1.0,
        1.0
    ],
    "recall_incomplete": [
        0.0,
        0.0
    ]
}
},
{
    "branch_percent": 50,
    "arm_condition": "parked",
    "method": "DT + S3",
    "threshold": 0.4499065998983014,
    "auroc": 1.0,
    "balanced_accuracy": 0.5,
    "accuracy": 0.3333333333333333,
    "recall_success": 1.0,
    "recall_incomplete": 0.0,
    "tn": 0,
    "fp": 8,
    "fn": 0,
    "tp": 4,
    "within_source_auroc": 1.0,
    "ci95_source_cluster": {
        "auroc": [
            1.0,
            1.0
        ],
        "balanced_accuracy": [
            0.5,
            0.5
        ],
        "recall_success": [
            1.0,
            1.0
        ],
        "recall_incomplete": [
            0.0,
            0.0
        ]
    }
},
{
    "branch_percent": 50,
    "arm_condition": "parked",
    "method": "DT + S5",
    "threshold": 0.4502469361783752,
    "auroc": 1.0,
    "balanced_accuracy": 0.5,
    "accuracy": 0.3333333333333333,
    "recall_success": 1.0,
    "recall_incomplete": 0.0,
    "tn": 0,
    "fp": 8,
    "fn": 0,
    "tp": 4,
    "within_source_auroc": 1.0,

```

```

"ci95_source_cluster": {
  "auroc": [
    1.0,
    1.0
  ],
  "balanced_accuracy": [
    0.5,
    0.5
  ],
  "recall_success": [
    1.0,
    1.0
  ],
  "recall_incomplete": [
    0.0,
    0.0
  ]
},
{
  "branch_percent": 50,
  "arm_condition": "parked",
  "method": "Always successful",
  "threshold": 0.5,
  "auroc": 0.5,
  "balanced_accuracy": 0.5,
  "accuracy": 0.3333333333333333,
  "recall_success": 1.0,
  "recall_incomplete": 0.0,
  "tn": 0,
  "fp": 8,
  "fn": 0,
  "tp": 4,
  "within_source_auroc": 0.5,
  "ci95_source_cluster": {
    "auroc": [
      0.5,
      0.5
    ],
    "balanced_accuracy": [
      0.5,
      0.5
    ],
    "recall_success": [
      1.0,
      1.0
    ],
    "recall_incomplete": [
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      0.0
    ]
  }
},
{
  "branch_percent": 50,
  "arm_condition": "parked",
  "method": "Robometer progress",
  "threshold": null,
  "auroc": 0.8125,
  "balanced_accuracy": null,

```

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"accuracy": null,
"recall_success": null,
"recall_incomplete": null,
"tn": null,
"fp": null,
"fn": null,
"tp": null,
"within_source_auroc": 0.875,
"ci95_source_cluster": {
  "auroc": [
    0.625,
    1.0
  ],
  "balanced_accuracy": null,
  "recall_success": null,
  "recall_incomplete": null
}
},
{
  "branch_percent": 50,
  "arm_condition": "parked",
  "method": "Robometer success",
  "threshold": 0.5,
  "auroc": 0.8125,
  "balanced_accuracy": 0.5,
  "accuracy": 0.6666666666666666,
  "recall_success": 0.0,
  "recall_incomplete": 1.0,
  "tn": 8,
  "fp": 0,
  "fn": 4,
  "tp": 0,
  "within_source_auroc": 0.75,
  "ci95_source_cluster": {
    "auroc": [
      0.25,
      1.0
    ],
    "balanced_accuracy": [
      0.5,
      0.5
    ],
    "recall_success": [
      0.0,
      0.0
    ],
    "recall_incomplete": [
      1.0,
      1.0
    ]
  }
},
{
  "branch_percent": 70,
  "arm_condition": "original",
  "method": "DINOv2",
  "threshold": 0.8599722981452942,
  "auroc": 1.0,
  "balanced_accuracy": 0.9,
  "accuracy": 0.9166666666666666,

```

```

"recall_success": 0.8,
"recall_incomplete": 1.0,
"tn": 7,
"fp": 0,
"fn": 1,
"tp": 4,
"within_source_auroc": 1.0,
"ci95_source_cluster": {
  "auroc": [
    1.0,
    1.0
  ],
  "balanced_accuracy": [
    0.7857142857142857,
    1.0
  ],
  "recall_success": [
    0.5714285714285714,
    1.0
  ],
  "recall_incomplete": [
    1.0,
    1.0
  ]
}
},
{
  "branch_percent": 70,
  "arm_condition": "original",
  "method": "DT",
  "threshold": -8.942145824432373,
  "auroc": 0.9714285714285714,
  "balanced_accuracy": 0.9285714285714286,
  "accuracy": 0.9166666666666666,
  "recall_success": 1.0,
  "recall_incomplete": 0.8571428571428571,
  "tn": 6,
  "fp": 1,
  "fn": 0,
  "tp": 5,
  "within_source_auroc": 0.875,
  "ci95_source_cluster": {
    "auroc": [
      0.7428571428571429,
      1.0
    ],
    "balanced_accuracy": [
      0.7,
      1.0
    ],
    "recall_success": [
      1.0,
      1.0
    ],
    "recall_incomplete": [
      0.4,
      1.0
    ]
  }
}
},

```

```

{
  "branch_percent": 70,
  "arm_condition": "original",
  "method": "Elapsed time",
  "threshold": 34.43333333333334,
  "auroc": 0.37142857142857144,
  "balanced_accuracy": 0.5,
  "accuracy": 0.5833333333333334,
  "recall_success": 0.0,
  "recall_incomplete": 1.0,
  "tn": 7,
  "fp": 0,
  "fn": 5,
  "tp": 0,
  "within_source_auroc": 0.5,
  "ci95_source_cluster": {
    "auroc": [
      0.3333333333333333,
      0.5
    ],
    "balanced_accuracy": [
      0.5,
      0.5
    ],
    "recall_success": [
      0.0,
      0.0
    ],
    "recall_incomplete": [
      1.0,
      1.0
    ]
  }
},
{
  "branch_percent": 70,
  "arm_condition": "original",
  "method": "S1",
  "threshold": -3.272519528865814,
  "auroc": 0.9714285714285714,
  "balanced_accuracy": 0.9,
  "accuracy": 0.9166666666666666,
  "recall_success": 0.8,
  "recall_incomplete": 1.0,
  "tn": 7,
  "fp": 0,
  "fn": 1,
  "tp": 4,
  "within_source_auroc": 1.0,
  "ci95_source_cluster": {
    "auroc": [
      0.9142857142857143,
      1.0
    ],
    "balanced_accuracy": [
      0.7857142857142857,
      1.0
    ],
    "recall_success": [
      0.5714285714285714,

```

```

    1.0
  ],
  "recall_incomplete": [
    1.0,
    1.0
  ]
}
},
{
  "branch_percent": 70,
  "arm_condition": "original",
  "method": "S3",
  "threshold": -3.2162272334098816,
  "auroc": 0.9714285714285714,
  "balanced_accuracy": 0.9,
  "accuracy": 0.9166666666666666,
  "recall_success": 0.8,
  "recall_incomplete": 1.0,
  "tn": 7,
  "fp": 0,
  "fn": 1,
  "tp": 4,
  "within_source_auroc": 1.0,
  "ci95_source_cluster": {
    "auroc": [
      0.8888888888888888,
      1.0
    ],
    "balanced_accuracy": [
      0.7857142857142857,
      1.0
    ],
    "recall_success": [
      0.5714285714285714,
      1.0
    ],
    "recall_incomplete": [
      1.0,
      1.0
    ]
  }
},
{
  "branch_percent": 70,
  "arm_condition": "original",
  "method": "S5",
  "threshold": -3.123962104320526,
  "auroc": 0.9428571428571428,
  "balanced_accuracy": 0.9,
  "accuracy": 0.9166666666666666,
  "recall_success": 0.8,
  "recall_incomplete": 1.0,
  "tn": 7,
  "fp": 0,
  "fn": 1,
  "tp": 4,
  "within_source_auroc": 1.0,
  "ci95_source_cluster": {
    "auroc": [
      0.8888888888888888,

```

```

    1.0
  ],
  "balanced_accuracy": [
    0.7857142857142857,
    1.0
  ],
  "recall_success": [
    0.5714285714285714,
    1.0
  ],
  "recall_incomplete": [
    1.0,
    1.0
  ]
}
},
{
  "branch_percent": 70,
  "arm_condition": "original",
  "method": "DT + S1",
  "threshold": 0.448230025164487,
  "auroc": 1.0,
  "balanced_accuracy": 0.9,
  "accuracy": 0.9166666666666666,
  "recall_success": 0.8,
  "recall_incomplete": 1.0,
  "tn": 7,
  "fp": 0,
  "fn": 1,
  "tp": 4,
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  "ci95_source_cluster": {
    "auroc": [
      1.0,
      1.0
    ],
    "balanced_accuracy": [
      0.7857142857142857,
      1.0
    ],
    "recall_success": [
      0.5714285714285714,
      1.0
    ],
    "recall_incomplete": [
      1.0,
      1.0
    ]
  }
},
{
  "branch_percent": 70,
  "arm_condition": "original",
  "method": "DT + S3",
  "threshold": 0.4499065998983014,
  "auroc": 1.0,
  "balanced_accuracy": 0.9,
  "accuracy": 0.9166666666666666,
  "recall_success": 0.8,
  "recall_incomplete": 1.0,

```

```

"tn": 7,
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"tp": 4,
"within_source_auroc": 1.0,
"ci95_source_cluster": {
  "auroc": [
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    1.0
  ],
  "balanced_accuracy": [
    0.7857142857142857,
    1.0
  ],
  "recall_success": [
    0.5714285714285714,
    1.0
  ],
  "recall_incomplete": [
    1.0,
    1.0
  ]
}
},
{
  "branch_percent": 70,
  "arm_condition": "original",
  "method": "DT + S5",
  "threshold": 0.4502469361783752,
  "auroc": 0.9714285714285714,
  "balanced_accuracy": 0.9,
  "accuracy": 0.9166666666666666,
  "recall_success": 0.8,
  "recall_incomplete": 1.0,
  "tn": 7,
  "fp": 0,
  "fn": 1,
  "tp": 4,
  "within_source_auroc": 1.0,
  "ci95_source_cluster": {
    "auroc": [
      0.9142857142857143,
      1.0
    ],
    "balanced_accuracy": [
      0.7857142857142857,
      1.0
    ],
    "recall_success": [
      0.5714285714285714,
      1.0
    ],
    "recall_incomplete": [
      1.0,
      1.0
    ]
  }
}
},
{
  "branch_percent": 70,

```

```

"arm_condition": "original",
"method": "Always successful",
"threshold": 0.5,
"auroc": 0.5,
"balanced_accuracy": 0.5,
"accuracy": 0.4166666666666667,
"recall_success": 1.0,
"recall_incomplete": 0.0,
"tn": 0,
"fp": 7,
"fn": 0,
"tp": 5,
"within_source_auroc": 0.5,
"ci95_source_cluster": {
  "auroc": [
    0.5,
    0.5
  ],
  "balanced_accuracy": [
    0.5,
    0.5
  ],
  "recall_success": [
    1.0,
    1.0
  ],
  "recall_incomplete": [
    0.0,
    0.0
  ]
},
{
  "branch_percent": 70,
  "arm_condition": "original",
  "method": "Robometer progress",
  "threshold": null,
  "auroc": 0.5428571428571428,
  "balanced_accuracy": null,
  "accuracy": null,
  "recall_success": null,
  "recall_incomplete": null,
  "tn": null,
  "fp": null,
  "fn": null,
  "tp": null,
  "within_source_auroc": 0.5,
  "ci95_source_cluster": {
    "auroc": [
      0.21875,
      0.78125
    ],
    "balanced_accuracy": null,
    "recall_success": null,
    "recall_incomplete": null
  }
},
{
  "branch_percent": 70,
  "arm_condition": "original",

```

```

"method": "Robometer success",
"threshold": 0.5,
"auroc": 0.8285714285714286,
"balanced_accuracy": 0.5,
"accuracy": 0.5833333333333334,
"recall_success": 0.0,
"recall_incomplete": 1.0,
"tn": 7,
"fp": 0,
"fn": 5,
"tp": 0,
"within_source_auroc": 0.875,
"ci95_source_cluster": {
  "auroc": [
    0.6571428571428571,
    1.0
  ],
  "balanced_accuracy": [
    0.5,
    0.5
  ],
  "recall_success": [
    0.0,
    0.0
  ],
  "recall_incomplete": [
    1.0,
    1.0
  ]
}
},
{
  "branch_percent": 70,
  "arm_condition": "parked",
  "method": "DIN0v2",
  "threshold": 0.8599722981452942,
  "auroc": 0.9444444444444444,
  "balanced_accuracy": 0.5,
  "accuracy": 0.5,
  "recall_success": 1.0,
  "recall_incomplete": 0.0,
  "tn": 0,
  "fp": 6,
  "fn": 0,
  "tp": 6,
  "within_source_auroc": 0.875,
  "ci95_source_cluster": {
    "auroc": [
      0.71875,
      1.0
    ],
    "balanced_accuracy": [
      0.5,
      0.5
    ],
    "recall_success": [
      1.0,
      1.0
    ],
    "recall_incomplete": [

```

```

    0.0,
    0.0
  ]
},
{
  "branch_percent": 70,
  "arm_condition": "parked",
  "method": "DT",
  "threshold": -8.942145824432373,
  "auroc": 0.6111111111111112,
  "balanced_accuracy": 0.5,
  "accuracy": 0.5,
  "recall_success": 1.0,
  "recall_incomplete": 0.0,
  "tn": 0,
  "fp": 6,
  "fn": 0,
  "tp": 6,
  "within_source_auroc": 0.875,
  "ci95_source_cluster": {
    "auroc": [
      0.4444444444444444,
      1.0
    ],
    "balanced_accuracy": [
      0.5,
      0.5
    ],
    "recall_success": [
      1.0,
      1.0
    ],
    "recall_incomplete": [
      0.0,
      0.0
    ]
  }
},
{
  "branch_percent": 70,
  "arm_condition": "parked",
  "method": "Elapsed time",
  "threshold": 34.43333333333334,
  "auroc": 0.4166666666666667,
  "balanced_accuracy": 0.4166666666666663,
  "accuracy": 0.4166666666666667,
  "recall_success": 0.6666666666666666,
  "recall_incomplete": 0.1666666666666666,
  "tn": 1,
  "fp": 5,
  "fn": 2,
  "tp": 4,
  "within_source_auroc": 0.5,
  "ci95_source_cluster": {
    "auroc": [
      0.3333333333333333,
      0.6666666666666666
    ],
    "balanced_accuracy": [

```

```

    0.3333333333333333,
    0.5
  ],
  "recall_success": [
    0.14285714285714285,
    1.0
  ],
  "recall_incomplete": [
    0.0,
    0.75
  ]
}
},
{
  "branch_percent": 70,
  "arm_condition": "parked",
  "method": "S1",
  "threshold": -3.272519528865814,
  "auroc": 1.0,
  "balanced_accuracy": 0.5,
  "accuracy": 0.5,
  "recall_success": 1.0,
  "recall_incomplete": 0.0,
  "tn": 0,
  "fp": 6,
  "fn": 0,
  "tp": 6,
  "within_source_auroc": 1.0,
  "ci95_source_cluster": {
    "auroc": [
      1.0,
      1.0
    ],
    "balanced_accuracy": [
      0.5,
      0.5
    ],
    "recall_success": [
      1.0,
      1.0
    ],
    "recall_incomplete": [
      0.0,
      0.0
    ]
  }
},
{
  "branch_percent": 70,
  "arm_condition": "parked",
  "method": "S3",
  "threshold": -3.2162272334098816,
  "auroc": 0.9722222222222222,
  "balanced_accuracy": 0.5,
  "accuracy": 0.5,
  "recall_success": 1.0,
  "recall_incomplete": 0.0,
  "tn": 0,
  "fp": 6,
  "fn": 0,

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"tp": 6,
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"ci95_source_cluster": {
  "auroc": [
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    1.0
  ],
  "balanced_accuracy": [
    0.5,
    0.5
  ],
  "recall_success": [
    1.0,
    1.0
  ],
  "recall_incomplete": [
    0.0,
    0.0
  ]
}
},
{
  "branch_percent": 70,
  "arm_condition": "parked",
  "method": "S5",
  "threshold": -3.123962104320526,
  "auroc": 1.0,
  "balanced_accuracy": 0.5,
  "accuracy": 0.5,
  "recall_success": 1.0,
  "recall_incomplete": 0.0,
  "tn": 0,
  "fp": 6,
  "fn": 0,
  "tp": 6,
  "within_source_auroc": 1.0,
  "ci95_source_cluster": {
    "auroc": [
      1.0,
      1.0
    ],
    "balanced_accuracy": [
      0.5,
      0.5
    ],
    "recall_success": [
      1.0,
      1.0
    ],
    "recall_incomplete": [
      0.0,
      0.0
    ]
  }
}
},
{
  "branch_percent": 70,
  "arm_condition": "parked",
  "method": "DT + S1",
  "threshold": 0.448230025164487,

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    "auroc": 0.8611111111111112,
    "balanced_accuracy": 0.5,
    "accuracy": 0.5,
    "recall_success": 1.0,
    "recall_incomplete": 0.0,
    "tn": 0,
    "fp": 6,
    "fn": 0,
    "tp": 6,
    "within_source_auroc": 0.875,
    "ci95_source_cluster": {
      "auroc": [
        0.625,
        1.0
      ],
      "balanced_accuracy": [
        0.5,
        0.5
      ],
      "recall_success": [
        1.0,
        1.0
      ],
      "recall_incomplete": [
        0.0,
        0.0
      ]
    }
  },
  {
    "branch_percent": 70,
    "arm_condition": "parked",
    "method": "DT + S3",
    "threshold": 0.4499065998983014,
    "auroc": 0.8611111111111112,
    "balanced_accuracy": 0.5,
    "accuracy": 0.5,
    "recall_success": 1.0,
    "recall_incomplete": 0.0,
    "tn": 0,
    "fp": 6,
    "fn": 0,
    "tp": 6,
    "within_source_auroc": 0.875,
    "ci95_source_cluster": {
      "auroc": [
        0.625,
        1.0
      ],
      "balanced_accuracy": [
        0.5,
        0.5
      ],
      "recall_success": [
        1.0,
        1.0
      ],
      "recall_incomplete": [
        0.0,
        0.0
      ]
    }
  }
}

```

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    ]
  },
  {
    "branch_percent": 70,
    "arm_condition": "parked",
    "method": "DT + S5",
    "threshold": 0.4502469361783752,
    "auroc": 0.8611111111111112,
    "balanced_accuracy": 0.5,
    "accuracy": 0.5,
    "recall_success": 1.0,
    "recall_incomplete": 0.0,
    "tn": 0,
    "fp": 6,
    "fn": 0,
    "tp": 6,
    "within_source_auroc": 0.875,
    "ci95_source_cluster": {
      "auroc": [
        0.625,
        1.0
      ],
      "balanced_accuracy": [
        0.5,
        0.5
      ],
      "recall_success": [
        1.0,
        1.0
      ],
      "recall_incomplete": [
        0.0,
        0.0
      ]
    }
  },
  {
    "branch_percent": 70,
    "arm_condition": "parked",
    "method": "Always successful",
    "threshold": 0.5,
    "auroc": 0.5,
    "balanced_accuracy": 0.5,
    "accuracy": 0.5,
    "recall_success": 1.0,
    "recall_incomplete": 0.0,
    "tn": 0,
    "fp": 6,
    "fn": 0,
    "tp": 6,
    "within_source_auroc": 0.5,
    "ci95_source_cluster": {
      "auroc": [
        0.5,
        0.5
      ],
      "balanced_accuracy": [
        0.5,
        0.5
      ]
    }
  }
}

```

```

    ],
    "recall_success": [
        1.0,
        1.0
    ],
    "recall_incomplete": [
        0.0,
        0.0
    ]
}
},
{
    "branch_percent": 70,
    "arm_condition": "parked",
    "method": "Robometer progress",
    "threshold": null,
    "auroc": 0.6388888888888888,
    "balanced_accuracy": null,
    "accuracy": null,
    "recall_success": null,
    "recall_incomplete": null,
    "tn": null,
    "fp": null,
    "fn": null,
    "tp": null,
    "within_source_auroc": 0.875,
    "ci95_source_cluster": {
        "auroc": [
            0.4444444444444444,
            1.0
        ],
        "balanced_accuracy": null,
        "recall_success": null,
        "recall_incomplete": null
    }
},
{
    "branch_percent": 70,
    "arm_condition": "parked",
    "method": "Robometer success",
    "threshold": 0.5,
    "auroc": 0.7222222222222222,
    "balanced_accuracy": 0.5,
    "accuracy": 0.5,
    "recall_success": 0.0,
    "recall_incomplete": 1.0,
    "tn": 6,
    "fp": 0,
    "fn": 6,
    "tp": 0,
    "within_source_auroc": 0.75,
    "ci95_source_cluster": {
        "auroc": [
            0.3142857142857143,
            1.0
        ],
        "balanced_accuracy": [
            0.5,
            0.5
        ]
    }
},

```

```

    "recall_success": [
      0.0,
      0.0
    ],
    "recall_incomplete": [
      1.0,
      1.0
    ]
  }
},
{
  "branch_percent": 90,
  "arm_condition": "original",
  "method": "DIN0v2",
  "threshold": 0.8599722981452942,
  "auroc": null,
  "balanced_accuracy": null,
  "accuracy": 0.3333333333333333,
  "recall_success": 0.3333333333333333,
  "recall_incomplete": null,
  "tn": 0,
  "fp": 0,
  "fn": 8,
  "tp": 4,
  "within_source_auroc": null,
  "ci95_source_cluster": {
    "auroc": null,
    "balanced_accuracy": null,
    "recall_success": [
      0.3333333333333333,
      0.3333333333333333
    ],
    "recall_incomplete": null
  }
},
{
  "branch_percent": 90,
  "arm_condition": "original",
  "method": "DT",
  "threshold": -8.942145824432373,
  "auroc": null,
  "balanced_accuracy": null,
  "accuracy": 0.9166666666666666,
  "recall_success": 0.9166666666666666,
  "recall_incomplete": null,
  "tn": 0,
  "fp": 0,
  "fn": 1,
  "tp": 11,
  "within_source_auroc": null,
  "ci95_source_cluster": {
    "auroc": null,
    "balanced_accuracy": null,
    "recall_success": [
      0.75,
      1.0
    ],
    "recall_incomplete": null
  }
},

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{
  "branch_percent": 90,
  "arm_condition": "original",
  "method": "Elapsed time",
  "threshold": 34.43333333333334,
  "auroc": null,
  "balanced_accuracy": null,
  "accuracy": 0.0,
  "recall_success": 0.0,
  "recall_incomplete": null,
  "tn": 0,
  "fp": 0,
  "fn": 12,
  "tp": 0,
  "within_source_auroc": null,
  "ci95_source_cluster": {
    "auroc": null,
    "balanced_accuracy": null,
    "recall_success": [
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      0.0
    ],
    "recall_incomplete": null
  }
},
{
  "branch_percent": 90,
  "arm_condition": "original",
  "method": "S1",
  "threshold": -3.272519528865814,
  "auroc": null,
  "balanced_accuracy": null,
  "accuracy": 0.3333333333333333,
  "recall_success": 0.3333333333333333,
  "recall_incomplete": null,
  "tn": 0,
  "fp": 0,
  "fn": 8,
  "tp": 4,
  "within_source_auroc": null,
  "ci95_source_cluster": {
    "auroc": null,
    "balanced_accuracy": null,
    "recall_success": [
      0.3333333333333333,
      0.3333333333333333
    ],
    "recall_incomplete": null
  }
},
{
  "branch_percent": 90,
  "arm_condition": "original",
  "method": "S3",
  "threshold": -3.2162272334098816,
  "auroc": null,
  "balanced_accuracy": null,
  "accuracy": 0.3333333333333333,
  "recall_success": 0.3333333333333333,
  "recall_incomplete": null,

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"tn": 0,
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"tp": 4,
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"ci95_source_cluster": {
  "auroc": null,
  "balanced_accuracy": null,
  "recall_success": [
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    0.3333333333333333
  ],
  "recall_incomplete": null
}
},
{
  "branch_percent": 90,
  "arm_condition": "original",
  "method": "S5",
  "threshold": -3.123962104320526,
  "auroc": null,
  "balanced_accuracy": null,
  "accuracy": 0.3333333333333333,
  "recall_success": 0.3333333333333333,
  "recall_incomplete": null,
  "tn": 0,
  "fp": 0,
  "fn": 8,
  "tp": 4,
  "within_source_auroc": null,
  "ci95_source_cluster": {
    "auroc": null,
    "balanced_accuracy": null,
    "recall_success": [
      0.3333333333333333,
      0.3333333333333333
    ],
    "recall_incomplete": null
  }
},
{
  "branch_percent": 90,
  "arm_condition": "original",
  "method": "DT + S1",
  "threshold": 0.448230025164487,
  "auroc": null,
  "balanced_accuracy": null,
  "accuracy": 0.3333333333333333,
  "recall_success": 0.3333333333333333,
  "recall_incomplete": null,
  "tn": 0,
  "fp": 0,
  "fn": 8,
  "tp": 4,
  "within_source_auroc": null,
  "ci95_source_cluster": {
    "auroc": null,
    "balanced_accuracy": null,
    "recall_success": [
      0.3333333333333333,

```

```

    0.3333333333333333
  ],
  "recall_incomplete": null
},
{
  "branch_percent": 90,
  "arm_condition": "original",
  "method": "DT + S3",
  "threshold": 0.4499065998983014,
  "auroc": null,
  "balanced_accuracy": null,
  "accuracy": 0.3333333333333333,
  "recall_success": 0.3333333333333333,
  "recall_incomplete": null,
  "tn": 0,
  "fp": 0,
  "fn": 8,
  "tp": 4,
  "within_source_auroc": null,
  "ci95_source_cluster": {
    "auroc": null,
    "balanced_accuracy": null,
    "recall_success": [
      0.3333333333333333,
      0.3333333333333333
    ],
    "recall_incomplete": null
  }
},
{
  "branch_percent": 90,
  "arm_condition": "original",
  "method": "DT + S5",
  "threshold": 0.4502469361783752,
  "auroc": null,
  "balanced_accuracy": null,
  "accuracy": 0.3333333333333333,
  "recall_success": 0.3333333333333333,
  "recall_incomplete": null,
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  "fp": 0,
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    "auroc": null,
    "balanced_accuracy": null,
    "recall_success": [
      0.3333333333333333,
      0.3333333333333333
    ],
    "recall_incomplete": null
  }
},
{
  "branch_percent": 90,
  "arm_condition": "original",
  "method": "Always successful",
  "threshold": 0.5,

```

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    "auroc": null,
    "balanced_accuracy": null,
    "accuracy": 1.0,
    "recall_success": 1.0,
    "recall_incomplete": null,
    "tn": 0,
    "fp": 0,
    "fn": 0,
    "tp": 12,
    "within_source_auroc": null,
    "ci95_source_cluster": {
      "auroc": null,
      "balanced_accuracy": null,
      "recall_success": [
        1.0,
        1.0
      ],
      "recall_incomplete": null
    }
  },
  {
    "branch_percent": 90,
    "arm_condition": "original",
    "method": "Robometer progress",
    "threshold": null,
    "auroc": null,
    "balanced_accuracy": null,
    "accuracy": null,
    "recall_success": null,
    "recall_incomplete": null,
    "tn": null,
    "fp": null,
    "fn": null,
    "tp": null,
    "within_source_auroc": null,
    "ci95_source_cluster": {
      "auroc": null,
      "balanced_accuracy": null,
      "recall_success": null,
      "recall_incomplete": null
    }
  },
  {
    "branch_percent": 90,
    "arm_condition": "original",
    "method": "Robometer success",
    "threshold": 0.5,
    "auroc": null,
    "balanced_accuracy": null,
    "accuracy": 0.08333333333333333,
    "recall_success": 0.08333333333333333,
    "recall_incomplete": null,
    "tn": 0,
    "fp": 0,
    "fn": 11,
    "tp": 1,
    "within_source_auroc": null,
    "ci95_source_cluster": {
      "auroc": null,
      "balanced_accuracy": null,

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    "recall_success": [
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      0.25
    ],
    "recall_incomplete": null
  },
  {
    "branch_percent": 90,
    "arm_condition": "parked",
    "method": "DINOv2",
    "threshold": 0.8599722981452942,
    "auroc": null,
    "balanced_accuracy": null,
    "accuracy": 1.0,
    "recall_success": 1.0,
    "recall_incomplete": null,
    "tn": 0,
    "fp": 0,
    "fn": 0,
    "tp": 12,
    "within_source_auroc": null,
    "ci95_source_cluster": {
      "auroc": null,
      "balanced_accuracy": null,
      "recall_success": [
        1.0,
        1.0
      ],
      "recall_incomplete": null
    }
  },
  {
    "branch_percent": 90,
    "arm_condition": "parked",
    "method": "DT",
    "threshold": -8.942145824432373,
    "auroc": null,
    "balanced_accuracy": null,
    "accuracy": 1.0,
    "recall_success": 1.0,
    "recall_incomplete": null,
    "tn": 0,
    "fp": 0,
    "fn": 0,
    "tp": 12,
    "within_source_auroc": null,
    "ci95_source_cluster": {
      "auroc": null,
      "balanced_accuracy": null,
      "recall_success": [
        1.0,
        1.0
      ],
      "recall_incomplete": null
    }
  },
  {
    "branch_percent": 90,
    "arm_condition": "parked",

```

```

"method": "Elapsed time",
"threshold": 34.43333333333334,
"auroc": null,
"balanced_accuracy": null,
"accuracy": 0.75,
"recall_success": 0.75,
"recall_incomplete": null,
"tn": 0,
"fp": 0,
"fn": 3,
"tp": 9,
"within_source_auroc": null,
"ci95_source_cluster": {
  "auroc": null,
  "balanced_accuracy": null,
  "recall_success": [
    0.25,
    1.0
  ],
  "recall_incomplete": null
},
},
{
  "branch_percent": 90,
  "arm_condition": "parked",
  "method": "S1",
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  "auroc": null,
  "balanced_accuracy": null,
  "accuracy": 1.0,
  "recall_success": 1.0,
  "recall_incomplete": null,
  "tn": 0,
  "fp": 0,
  "fn": 0,
  "tp": 12,
  "within_source_auroc": null,
  "ci95_source_cluster": {
    "auroc": null,
    "balanced_accuracy": null,
    "recall_success": [
      1.0,
      1.0
    ],
    "recall_incomplete": null
  }
},
{
  "branch_percent": 90,
  "arm_condition": "parked",
  "method": "S3",
  "threshold": -3.2162272334098816,
  "auroc": null,
  "balanced_accuracy": null,
  "accuracy": 1.0,
  "recall_success": 1.0,
  "recall_incomplete": null,
  "tn": 0,
  "fp": 0,
  "fn": 0,

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"tp": 12,
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"ci95_source_cluster": {
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  "balanced_accuracy": null,
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    1.0
  ],
  "recall_incomplete": null
},
},
{
  "branch_percent": 90,
  "arm_condition": "parked",
  "method": "S5",
  "threshold": -3.123962104320526,
  "auroc": null,
  "balanced_accuracy": null,
  "accuracy": 1.0,
  "recall_success": 1.0,
  "recall_incomplete": null,
  "tn": 0,
  "fp": 0,
  "fn": 0,
  "tp": 12,
  "within_source_auroc": null,
  "ci95_source_cluster": {
    "auroc": null,
    "balanced_accuracy": null,
    "recall_success": [
      1.0,
      1.0
    ],
    "recall_incomplete": null
  }
},
{
  "branch_percent": 90,
  "arm_condition": "parked",
  "method": "DT + S1",
  "threshold": 0.448230025164487,
  "auroc": null,
  "balanced_accuracy": null,
  "accuracy": 1.0,
  "recall_success": 1.0,
  "recall_incomplete": null,
  "tn": 0,
  "fp": 0,
  "fn": 0,
  "tp": 12,
  "within_source_auroc": null,
  "ci95_source_cluster": {
    "auroc": null,
    "balanced_accuracy": null,
    "recall_success": [
      1.0,
      1.0
    ],
    "recall_incomplete": null
  }
}

```

```

    }
  },
  {
    "branch_percent": 90,
    "arm_condition": "parked",
    "method": "DT + S3",
    "threshold": 0.4499065998983014,
    "auroc": null,
    "balanced_accuracy": null,
    "accuracy": 1.0,
    "recall_success": 1.0,
    "recall_incomplete": null,
    "tn": 0,
    "fp": 0,
    "fn": 0,
    "tp": 12,
    "within_source_auroc": null,
    "ci95_source_cluster": {
      "auroc": null,
      "balanced_accuracy": null,
      "recall_success": [
        1.0,
        1.0
      ],
      "recall_incomplete": null
    }
  },
  {
    "branch_percent": 90,
    "arm_condition": "parked",
    "method": "DT + S5",
    "threshold": 0.4502469361783752,
    "auroc": null,
    "balanced_accuracy": null,
    "accuracy": 1.0,
    "recall_success": 1.0,
    "recall_incomplete": null,
    "tn": 0,
    "fp": 0,
    "fn": 0,
    "tp": 12,
    "within_source_auroc": null,
    "ci95_source_cluster": {
      "auroc": null,
      "balanced_accuracy": null,
      "recall_success": [
        1.0,
        1.0
      ],
      "recall_incomplete": null
    }
  },
  {
    "branch_percent": 90,
    "arm_condition": "parked",
    "method": "Always successful",
    "threshold": 0.5,
    "auroc": null,
    "balanced_accuracy": null,
    "accuracy": 1.0,

```

```

"recall_success": 1.0,
"recall_incomplete": null,
"tn": 0,
"fp": 0,
"fn": 0,
"tp": 12,
"within_source_auroc": null,
"ci95_source_cluster": {
  "auroc": null,
  "balanced_accuracy": null,
  "recall_success": [
    1.0,
    1.0
  ],
  "recall_incomplete": null
},
},
{
  "branch_percent": 90,
  "arm_condition": "parked",
  "method": "Robometer progress",
  "threshold": null,
  "auroc": null,
  "balanced_accuracy": null,
  "accuracy": null,
  "recall_success": null,
  "recall_incomplete": null,
  "tn": null,
  "fp": null,
  "fn": null,
  "tp": null,
  "within_source_auroc": null,
  "ci95_source_cluster": {
    "auroc": null,
    "balanced_accuracy": null,
    "recall_success": null,
    "recall_incomplete": null
  }
},
{
  "branch_percent": 90,
  "arm_condition": "parked",
  "method": "Robometer success",
  "threshold": 0.5,
  "auroc": null,
  "balanced_accuracy": null,
  "accuracy": 0.0,
  "recall_success": 0.0,
  "recall_incomplete": null,
  "tn": 0,
  "fp": 0,
  "fn": 12,
  "tp": 0,
  "within_source_auroc": null,
  "ci95_source_cluster": {
    "auroc": null,
    "balanced_accuracy": null,
    "recall_success": [
      0.0,
      0.0
    ]
  }
}

```

```
    ],  
    "recall_incomplete": null  
  }  
}  
}
```